Pratham P N

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Education

Sahyadri College of Engineering and Management Mangaluru, Karnataka
Bachelor of Engineering in Artificial Intelligence and Machine Learning *July 2023 – July 2027*
CGPA: 9.22/10 — Full-time

Sharada PU College Kodialbail, Mangaluru, Karnataka
12th / PUC — Score: 92.8% *2023*
Full-time

Sri Sharadambha High School Sheni, Kasaragod, Kerala
10th / SSLC — Score: 100% *2021*
Full-time

Strong Subjects

Data Structures and Algorithms • Object-Oriented Programming • Database Management Systems • Operating Systems • Computer Networks • Artificial Intelligence • Machine Learning

Technical Skills

Programming Languages: C, C++, Java, Python, SQL, JavaScript

Frontend: HTML5, CSS3, React.js, Bootstrap, Tailwind CSS

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL

AI / Computer Vision: OpenCV, MediaPipe, Machine Learning

Developer Tools: Git, GitHub, Docker, VS Code, Vercel

Internship Experience

- **Hack Harbor 2.0 – C Programming Internship** Sahyadri College of Engineering and Management

  Completed a programming-focused internship involving C programming, problem solving, and practical software development. Worked collaboratively on a Financial Management application and participated in the associated hackathon and ideation activities.

Projects

- **AI-Powered Fitness Coach** Python, OpenCV, MediaPipe, Machine Learning

  Developed an AI-powered fitness coaching application that analyzes exercise posture using computer vision and pose estimation. Provides real-time feedback to improve exercise form and workout performance. The project is being extended toward AI-assisted exercise analysis and posture correction.

- **AI Pose Data Extraction System** Python, OpenCV, MediaPipe, Pandas

  Developed a preprocessing system that converts exercise videos into normalized CSV datasets using OpenCV and MediaPipe. Extracts human pose landmarks and generates machine-learning-ready datasets for exercise analysis and model development.

- **Exam Room Allocation System** Flutter, MySQL

  Developed an exam room allocation system to automate seating arrangements and student allocation. The system manages room assignments, reduces manual effort, minimizes examination malpractice, and provides administrative controls for room management and student lookup.

- **Non-Playable Characters (NPCs): A Multidisciplinary Review**

  Conducting a comprehensive survey of Non-Playable Characters across games, simulations, and animated media. Reviewed more than 80 research papers published between 2015 and 2025 to study NPC architectures, artificial intelligence techniques, behavioral models, research trends, and future directions.

Certifications

- Getting Started with Amazon Web Services (AWS) – Infosys Springboard
- Software Engineering – Infosys Springboard

Leadership & Activities

- Selected among the **Top 50 Teams** to participate in the **HackFest Hackathon**, Nitte.
- Participated in the **Aakruthi Hackathon**, Canara Engineering College, Benjanapadavu.
- Currently serving as **Joint Treasurer** of the Department Student Association.
- Volunteered in **DevHost** and multiple college technical events, supporting event planning and coordination.
- Assisted in organizing placement drives and departmental technical events.
- Learning multiple languages through **Duolingo**, maintaining a **470+ day learning streak**.

Interests

Artificial Intelligence • Full-Stack Development • Competitive Programming • Computer Vision • Language Learning